

Yuancheng (Remo) Shen

370 Jay St, Brooklyn, NY 11201, United States of America

☎ +1-201-204-2218 ✉ ys6345@nyu.edu 🌐 remoshen.github.io 📁 github.com/remoshen

EDUCATION

- **New York University** New York, United States
Ph.D. - Computer Science
Advisor: Prof. Robert Krueger
Sep. 2024 - present
- **Shandong University** Shandong, China
Master - Computer Science and Technology
Advisor: Prof. Yunhai Wang
Sep. 2021 - Jun. 2024
Courses: Human-Computer Interaction, Interactive Data Analysis System, Artificial Intelligence, Machine Learning
- **Jiangsu University of Science and Technology** Jiangsu, China
Bachelor - Computer Science and Technology
Sep. 2017 - Jun. 2021
GPA: Official: 3.85; Major: 3.95; Ranking: 2/102
Courses: Operating Systems, Data Structures, Analysis Of Algorithms, Computer Graphics, Networking, Databases

PUBLICATIONS

- [P1] **Y. Shen**, C. Kalsi, P. Motarwar, D. Annuar, J. Stimes, S. Sterling, S. Qiu, I. Illa-Bochaca, I. Osman, D. Fenyo, G. J. Baker, E. Hernando, K. Ruggles, A. Tsigirgos, and R. Krueger. CellGalaxy: Multiscale Exploration of Biomedical Image Embeddings. In *IEEE VIS, 2026* (In process).
- [P2] C. Kalsi, H. Fathollahian, L. Reichman, **Y. Shen**, S. Gaupp, A. Szabó, A. Rodríguez, T. Vallius, M. Rogava, G. E. Marai, M. Krone, S. Boorboor, and R. Krueger. BioSET: Interactive Visual Analysis of Biomarker Co-localization in Large Multi-Volume Tissue Data. In *IEEE VIS, 2026* (In process).
- [P3] **Yuancheng Shen**, Yue Zhao, Yunhai Wang, Tong Ge, Haoyan Shi and Bongshin Lee. Authoring Data-Driven Chart Animations through Direct Manipulation. In *IEEE Transactions on Visualization and Computer Graphics, 2024*. (DOI: 10.1109/TVCG.2024.3491504).
- [P4] C. Kalsi, **Y. Shen**, S. Gaupp, L. Reichmann, M. Rogava, M. Krone, S. Boorboor, and R. Krueger. BioSET - Biomarker-based Spatial co-Expression analysis in Tumor environments. In *Bio+MedVis Challenge @ IEEE VIS, 2025* (Award of Excellence).
- [P5] Tao Dai, Qi Wang, **Yuancheng Shen**, Shang Gao. SwinVision: detecting small objects in low-light environments. In *IEEE Access, 2025* (DOI: 10.1109/ACCESS.2025.3548151).
- [P6] **Yuancheng Shen**, Rui Ban, Xin Chen, Runduo Hua, Yunhai Wang. (2023). Anomaly Detection Algorithm for Network Device Configuration Based on Configuration Statement Tree. In *Computer Science*, vol. 50, no. 11A, pp. 230200128-10, 2023.

RESEARCH EXPERIENCE

- **3D Cellular Interaction Visualization for Multiplexed Tissue Imaging** New York University
Collaborator | Advisor: Robert Krueger
May. 2025 - Sep. 2025
 - **Description:** Designed a scalable visual analytics workflow for CyCIF melanoma imaging datasets (70 channels, multi-resolution pyramid, up to 1.48TB). The system supports large-scale identification and visualization of cell-cell interactions based on spatially co-located protein markers.
 - **Contribution:** Designed a scalable overview-to-detail visualization framework combining multi-resolution volume rendering, surface-based ROI exploration, and contextual biomarker querying. Implemented GPU-accelerated co-localization detection, semantic zooming for multi-scale exploration, and integrated semantic explanations to interpret spatial expression patterns in collaboration with biomedical experts from Northwell Health, NY.
- **Multi-Scale Visualization of Cellular Features in Cancerous Image Data** New York University
Student Leader | Advisor: Robert Krueger
Nov. 2024 - Present
 - **Description:** Design an interactive visual analytics tool for multiplexed cancer tissue imaging, supporting dynamic marker composition and feature-based cell embedding to reveal spatial and expression-driven tissue structures.
 - **Contribution:** Develop coordinated views integrating spatial maps, expression scatterplots, and dendrograms; implemented scalable hierarchical clustering, semantic zooming, and progressive filtering to support multi-scale analysis and real-time performance in large-scale datasets.
- **Live-Cell Dynamic Cell Imaging Visual Analytics** New York University
Collaborator | Advisor: Robert Krueger
Dec. 2024 - Present
 - **Description:** Develop a visualization system for analyzing live-cell imaging data, focusing on the dynamic interactions between immune and cancer cells using long-term 2D time-lapse microscopy.
 - **Contribution:** Implement interactive visualizations of cell trajectories to encode movement, interactions, and state transitions; Designed temporal filters and annotation tools to highlight immune killing events; Integrat trajectory data with biomarker expression and lineage tracking to support biological interpretation; Address challenges such as tracking noise and temporal resolution via adaptive smoothing and hierarchical time-window techniques.

- Authoring Data-Driven Chart Animations through Direct Manipulation [P1][Link]** Shandong University
Student Leader | Advisor: Yunhai Wang and Bongshin Lee Oct. 2023 - Jun. 2024
 - Description:** The research concentrates on an intuitive tool that empowers users without programming skills to author expressive chart animations through visual language, interactive editing, and smart recommendation strategies.
 - Contribution:** Researched data animation syntax and tools; Proposed and implemented innovative ideas in consultation with two advisors; Took responsibility for paper writing and illustrations.
 - Achievement:** Developed an interactive tool based on Canis syntax, enabling users to author data-driven chart animations with ease; Written a research paper.
- SwinVision: Detecting Small Objects in Low-light Environments** Shandong University
Collaborator | Advisor: Shang Gao Jan. 2024 - Sep. 2024
 - Description:** The research focuses on a Swin Transformer-based framework for small object detection, integrating feature enhancement and specialized modules to improve accuracy and efficiency in low-light environments.
 - Contribution:** Proposed innovative ideas and models for enhancing small object detection in low-light environments, ensuring a balanced approach between accuracy and efficiency; Contributed extensively to writing and refining the manuscript, with the majority of the content authored by me

SELECTED HONORS AND AWARDS

- NYU School of Engineering (SOE) PhD Fellowship, New York University Sep. 2024
- Outstanding Thesis Award, Shandong University Jun. 2024
- Outstanding Graduates Award, Shandong University Jun. 2024
- Postgraduate Excellent Student Award Fund, Shandong University Oct. 2021
- Outstanding Thesis Award, Jiangsu University of Science and Technology Jun. 2021
- Outstanding Graduates Award, Jiangsu University of Science and Technology Jun. 2021
- 1st Prize Scholarship, Jiangsu University of Science and Technology Oct. 2019
- 1st Prize in Higher Mathematics Competition, Jiangsu Aug. 2018

REVIEWS AND COMMUNITY WORK

- Reviewer:** Expert Systems With Applications; PeerJ Computer Science

ACADEMIC ENGAGEMENTS

- International Conference on Geometric Modeling and Processing** Qingdao, China
Participated in the event, received experts and scholars, and volunteered for other conference services Jun. 2024
- The Geometric Design and Computing Conference** Qingdao, China
Participated in the event, received experts and scholars, and volunteered for other conference services Aug. 2022
- The China Visualization and Visual Analytics Conference** Xining, China
Participated in the event Jul. 2022

SKILLS

- Programming:** Python, TypeScript, JavaScript, C++, SQL, R
- Machine Learning:** PyTorch, NumPy, Pandas
- Visualization & Tools:** WebGL, VTK.js, D3.js, Git, Linux, LaTeX, Adobe Illustrator